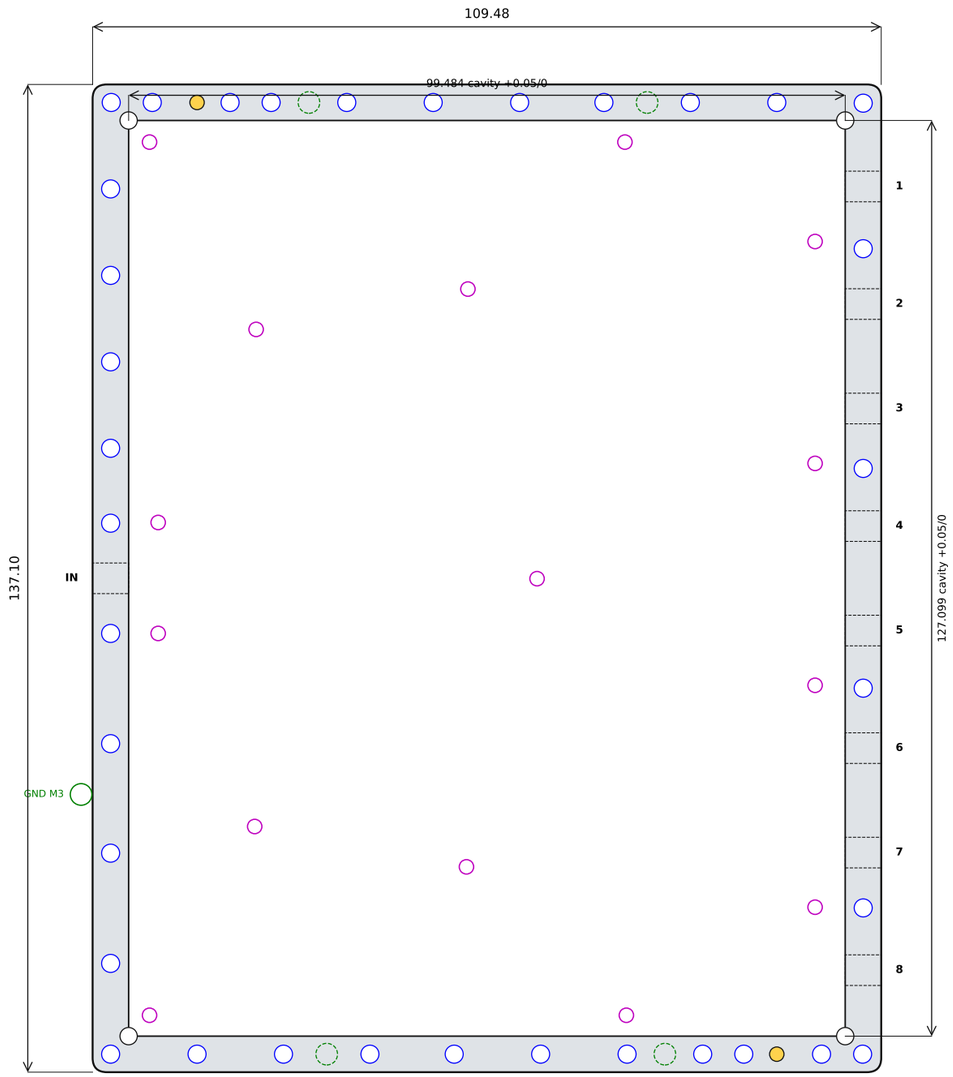
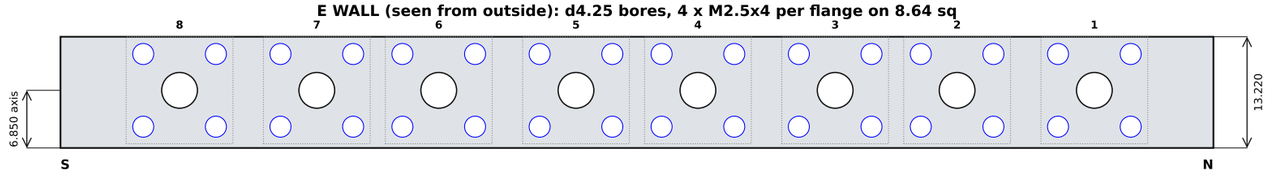
BASE CRL-XB-PD8-03-01 - TOP VIEW

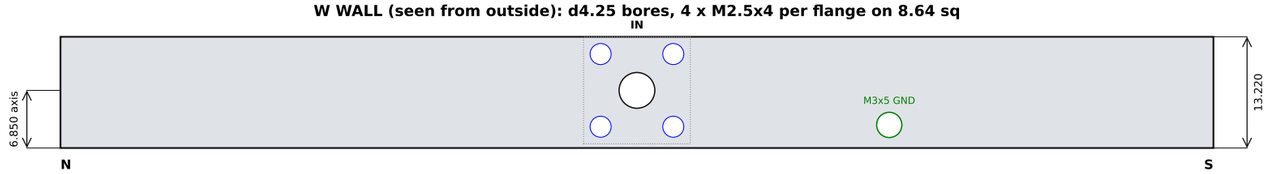


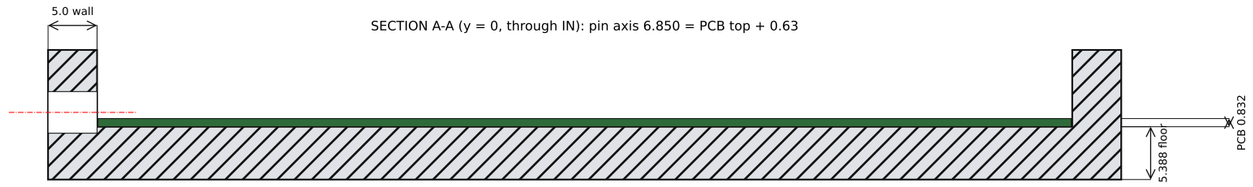
blue M2.5 lid taps (35) magenta M2 PCB taps (15) yellow dowel d2 (2) green dashed M3 mounting taps from below (4)
dashed: d4.25 connector bores through the wall. Corner reliefs d2.4. All coordinates in the hole table (sheet 4).

E WALL (seen from outside): d4.25 bores, 4 x M2.5x4 per flange on 8.64 sq



W WALL (seen from outside): d4.25 bores, 4 x M2.5x4 per flange on 8.64 sq





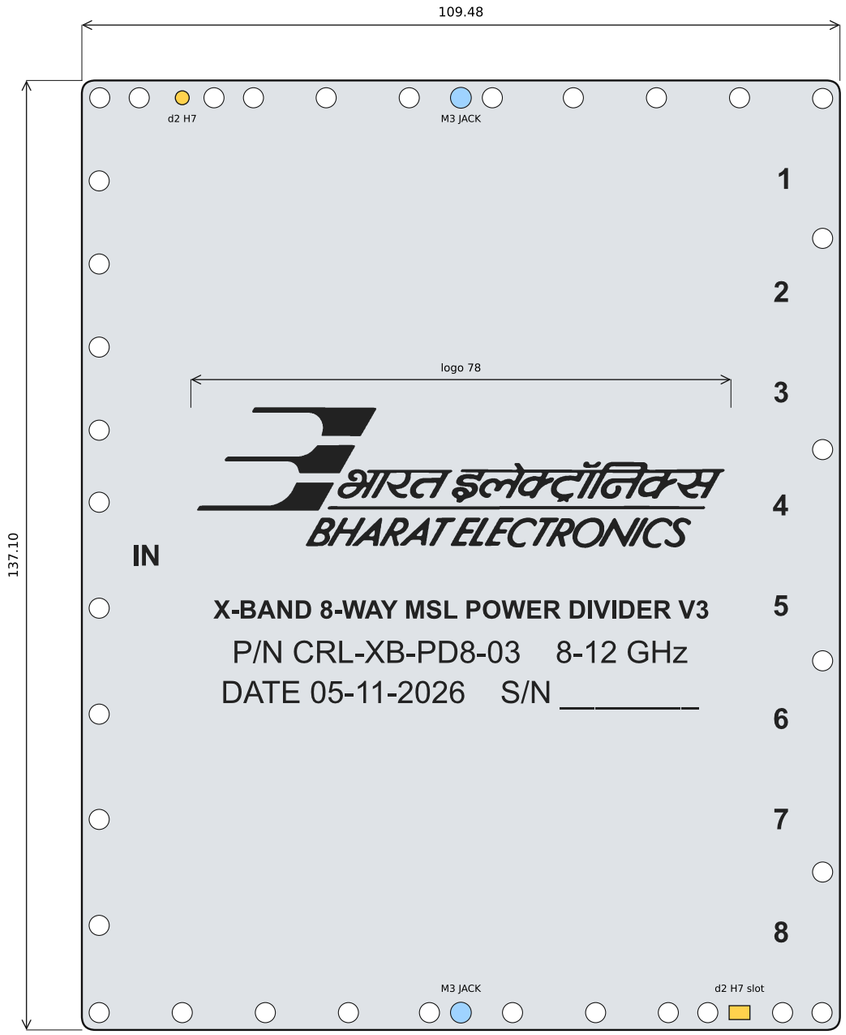
NOTES - BASE
1. Material Aluminium EN AW-6061-T6. Finish: Chromate conversion MIL-DTL-5541 Type II Class 3, clear (conductive), all over.
2. Tolerances ISO 2768-mK. Cavity, bores and dowel holes +/-0.02.
3. Lid seat (wall top) flat 0.03, Ra 0.8. Cavity floor flat 0.03 (PCB ground contact).
4. Connector bores d4.25 +0.02/0 through, coaxial with the flange tap pattern 8.64 sq.
5. Corners of the cavity: R1.0 with d2.4 reliefs (square PCB corners).
6. Break edges: base bottom 0.5 x 45, vertical outside edges R2, all others 0.2 max.
7. Engrave 0.3 deep on the W wall: RF IN, GND. Fill white (optional).
8. Press 2 x ISO 8734 2 m6 x 8 dowels 5 deep into the wall top.
9. 3D model: CRL-XB-PD8-03_BASE.step (governs where this sheet is silent).

BEL - CRL
ANTENNA / RFM LAB

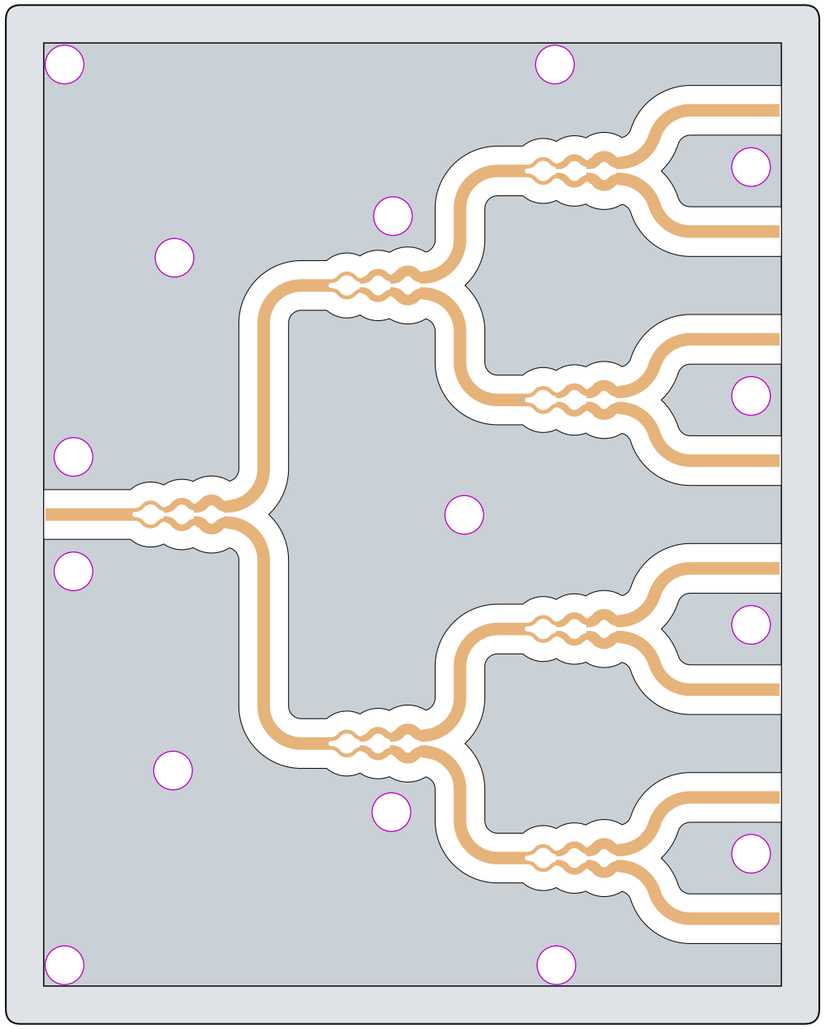
X-BAND 8-WAY MSL POWER DIVIDER V3
BASE - machining

PART	DWG	REV	SHEET
CRL-XB-PD8-03-01	CRL-XB-PD8-03-DWG	V3	1 / 4
DATE	UNITS	PROJ.	TOL.
05-11-2026	mm	1st angle	ISO 2768-mK

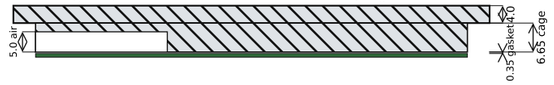
LID CRL-XB-PD8-03-02 - TOP: engrave 0.3 deep (artwork: dxf_src/lid_engraving.dxf)



LID - CAGE (underside, shown in the top-view frame): grey = face 0.35 above PCB, white = channels 4.65 deep channels 2.5 mm from the copper (orange = board copper for reference), magenta = d5.2 x 2.25 head pockets



SECTION B-B (y = 0) lid on base: channel over the IN line, cage face on the gasket



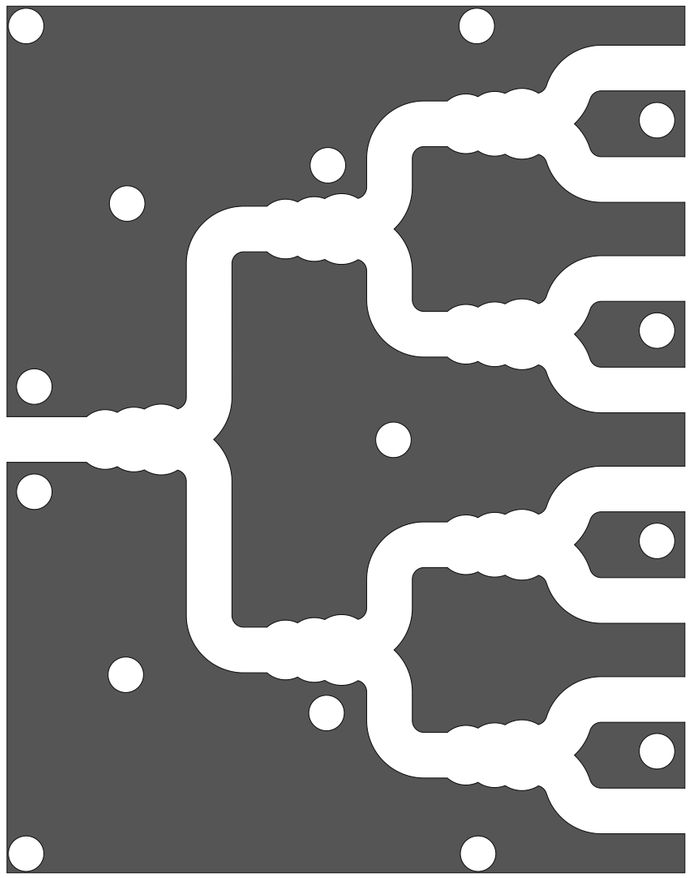
NOTES - LID
1. Material / finish as the base.
2. Cage face flat 0.03 and parallel to the lid seat 0.03; Ra 1.6.
3. Channels: D2 end mill, R1 min, depth 4.65 +0.05/0 from the face. Outline: dxf_src/lid_channels.dxf.
4. Engraving 0.3 deep: laser or D0.2 V-cutter (logo strokes 0.3). Text: logo, title, P/N, DATE 05-11-2026, S/N field, IN / 1-8 port labels.
5. Lid holes: 35 x d2.9 for M2.5x8 ISO 7380; d2 H7 hole + slot; 2 x M3 jacking taps through.
6. Break edges: top 0.6 x 45, R2.
7. Model: CRL-XB-PD8-03_LID.step.

BEL - CRL
ANTENNA / RFM LAB

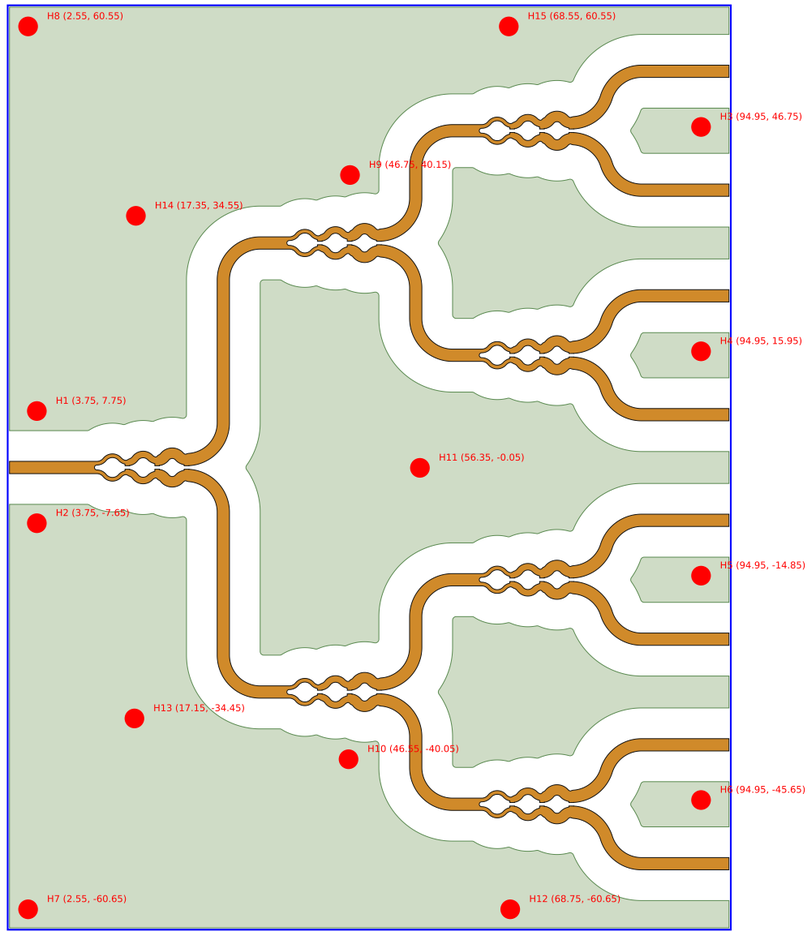
X-BAND 8-WAY MSL POWER DIVIDER V3
LID with channel cage - machining + engraving

PART	DWG	REV	SHEET
CRL-XB-PD8-03-02	CRL-XB-PD8-03-DWG	V3	2 / 4
DATE	UNITS	PROJ.	TOL.
05-11-2026	mm	1st angle	ISO 2768-mK

GASKET CRL-XB-PD8-03-03: 0.50 mm conductive elastomer, die-cut from dxf_src/gasket.dxf (top-view frame)



PCB V3: 15 x M2 PTH d2.40 finished (customer DXF frame) file: 01_pcb_v3/..._V3.dxf, layer MOUNT_HOLES_PTH



ASSEMBLY

1. Fit the PCB V3 in the base (copper side up, IN to the W wall). 15 x M2x5 ISO 4762 A2, 0.20 N m, centre outward.
2. Connectors 23_SMA-50-0-272/133_N: spigot into the bore, pin on the trace. Fit outputs 2-3, 4-5, 6-7 in order (flanges 1.8 apart). 4 x M2.5x5 ISO 4762 each, 0.35 N m. Solder each pin to its launch.
3. Clean flux. Lay the 0.50 mm gasket on the board (aligned to the channels) or bond it to the cage face with conductive adhesive.
4. Lid on the 2 dowels; 35 x M2.5x8 ISO 7380, 0.35 N m, cross pattern.
5. Removal: 2 x M3 jack screws through the lid. Never pry.
6. Test: S-parameters 8-12 GHz, all ports; lid fitted.

BEL - CRL

ANTENNA / RFM LAB

X-BAND 8-WAY MSL POWER DIVIDER V3

GASKET, PCB V3 holes, ASSEMBLY

PART	DWG	REV	SHEET
CRL-XB-PD8-03-03 / PCB	CRL-XB-PD8-03-DWG	V3	3 / 4
DATE	UNITS	PROJ.	TOL.
05-11-2026	mm	1st angle	ISO 2768-mK

HOLE TABLE - datum: outside SW corner of the base, bottom face. X east, Y north, Z up (Z = face the feature starts from). Also as 05_docs/CRL-XB-PD8-03_HOLE_TABLE.csv.

part	face	feature	X	Y	Z
BASE	top (wall)	M2.5 tap 4.6 deep	106.900	2.500	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	101.200	2.500	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	90.400	2.500	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	84.700	2.500	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	74.200	2.500	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	62.200	2.500	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	50.200	2.500	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	38.500	2.500	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	26.500	2.500	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	14.500	2.500	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	2.500	2.500	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	106.984	134.500	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	106.984	114.300	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	106.984	83.800	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	106.984	53.300	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	106.984	22.800	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	2.584	134.598	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	8.284	134.598	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	19.084	134.598	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	24.784	134.598	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	35.284	134.598	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	47.284	134.598	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	59.284	134.598	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	70.984	134.598	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	82.984	134.598	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	94.984	134.598	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	2.500	15.098	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	2.500	30.398	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	2.500	45.598	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	2.500	60.898	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	2.500	76.198	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	2.500	86.598	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	2.500	98.598	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	2.500	110.598	13.220
BASE	top (wall)	M2.5 tap 4.6 deep	2.500	122.598	13.220
BASE	top (floor)	M2 tap 4.5 deep	9.102	76.299	5.388
BASE	top (floor)	M2 tap 4.5 deep	9.102	60.899	5.388
BASE	top (floor)	M2 tap 4.5 deep	100.302	115.299	5.388
BASE	top (floor)	M2 tap 4.5 deep	100.302	84.499	5.388
BASE	top (floor)	M2 tap 4.5 deep	100.302	53.699	5.388
BASE	top (floor)	M2 tap 4.5 deep	100.302	22.899	5.388
BASE	top (floor)	M2 tap 4.5 deep	7.902	7.899	5.388
BASE	top (floor)	M2 tap 4.5 deep	7.902	129.099	5.388
BASE	top (floor)	M2 tap 4.5 deep	52.102	108.699	5.388
BASE	top (floor)	M2 tap 4.5 deep	51.902	28.499	5.388
BASE	top (floor)	M2 tap 4.5 deep	61.702	68.499	5.388
BASE	top (floor)	M2 tap 4.5 deep	74.102	7.899	5.388
BASE	top (floor)	M2 tap 4.5 deep	22.502	34.099	5.388
BASE	top (floor)	M2 tap 4.5 deep	22.702	103.099	5.388
BASE	top (floor)	M2 tap 4.5 deep	73.902	129.099	5.388
BASE	bottom	M3 tap 6.0 deep	32.500	2.500	0.000
BASE	bottom	M3 tap 6.0 deep	79.450	2.500	0.000
BASE	bottom	M3 tap 6.0 deep	30.034	134.598	0.000
BASE	bottom	M3 tap 6.0 deep	76.984	134.598	0.000
BASE	top (wall)	d2 press fit for 2 m6 dowel, 5.0 deep	14.500	134.598	13.220
BASE	top (wall)	d2 press fit for 2 m6 dowel, 5.0 deep	94.984	2.500	13.220
BASE	W wall	d4.25 bore through (spigot), port IN	0.000	68.549	6.850
BASE	W wall	M2.5 tap 4.0 deep (flange IN)	0.000	64.229	2.530
BASE	W wall	M2.5 tap 4.0 deep (flange IN)	0.000	64.229	11.170
BASE	W wall	M2.5 tap 4.0 deep (flange IN)	0.000	72.869	2.530
BASE	W wall	M2.5 tap 4.0 deep (flange IN)	0.000	72.869	11.170
BASE	E wall	d4.25 bore through (spigot), port 1	109.484	122.939	6.850
BASE	E wall	M2.5 tap 4.0 deep (flange 1)	109.484	118.619	2.530
BASE	E wall	M2.5 tap 4.0 deep (flange 1)	109.484	118.619	11.170
BASE	E wall	M2.5 tap 4.0 deep (flange 1)	109.484	127.259	2.530
BASE	E wall	M2.5 tap 4.0 deep (flange 1)	109.484	127.259	11.170
BASE	E wall	d4.25 bore through (spigot), port 2	109.484	106.619	6.850
BASE	E wall	M2.5 tap 4.0 deep (flange 2)	109.484	102.299	2.530
BASE	E wall	M2.5 tap 4.0 deep (flange 2)	109.484	102.299	11.170
BASE	E wall	M2.5 tap 4.0 deep (flange 2)	109.484	110.939	2.530
BASE	E wall	M2.5 tap 4.0 deep (flange 2)	109.484	110.939	11.170
BASE	E wall	d4.25 bore through (spigot), port 3	109.484	92.119	6.850
BASE	E wall	M2.5 tap 4.0 deep (flange 3)	109.484	87.799	2.530
BASE	E wall	M2.5 tap 4.0 deep (flange 3)	109.484	87.799	11.170
BASE	E wall	M2.5 tap 4.0 deep (flange 3)	109.484	96.439	2.530
BASE	E wall	M2.5 tap 4.0 deep (flange 3)	109.484	96.439	11.170
BASE	E wall	d4.25 bore through (spigot), port 4	109.484	75.799	6.850
BASE	E wall	M2.5 tap 4.0 deep (flange 4)	109.484	71.479	2.530
BASE	E wall	M2.5 tap 4.0 deep (flange 4)	109.484	80.119	2.530
BASE	E wall	M2.5 tap 4.0 deep (flange 4)	109.484	80.119	11.170
BASE	E wall	d4.25 bore through (spigot), port 5	109.484	61.299	6.850
BASE	E wall	M2.5 tap 4.0 deep (flange 5)	109.484	56.979	2.530
BASE	E wall	M2.5 tap 4.0 deep (flange 5)	109.484	56.979	11.170
BASE	E wall	M2.5 tap 4.0 deep (flange 5)	109.484	65.619	2.530
BASE	E wall	M2.5 tap 4.0 deep (flange 5)	109.484	65.619	11.170
BASE	E wall	d4.25 bore through (spigot), port 6	109.484	44.979	6.850
BASE	E wall	M2.5 tap 4.0 deep (flange 6)	109.484	40.659	2.530
BASE	E wall	M2.5 tap 4.0 deep (flange 6)	109.484	40.659	11.170
BASE	E wall	M2.5 tap 4.0 deep (flange 6)	109.484	49.299	2.530
BASE	E wall	M2.5 tap 4.0 deep (flange 6)	109.484	49.299	11.170
BASE	E wall	d4.25 bore through (spigot), port 7	109.484	30.479	6.850
BASE	E wall	M2.5 tap 4.0 deep (flange 7)	109.484	26.159	2.530
BASE	E wall	M2.5 tap 4.0 deep (flange 7)	109.484	26.159	11.170
BASE	E wall	M2.5 tap 4.0 deep (flange 7)	109.484	34.799	2.530
BASE	E wall	M2.5 tap 4.0 deep (flange 7)	109.484	34.799	11.170
BASE	E wall	d4.25 bore through (spigot), port 8	109.484	14.159	6.850
BASE	E wall	M2.5 tap 4.0 deep (flange 8)	109.484	9.839	2.530
BASE	E wall	M2.5 tap 4.0 deep (flange 8)	109.484	9.839	11.170
BASE	E wall	M2.5 tap 4.0 deep (flange 8)	109.484	18.479	2.530
BASE	E wall	M2.5 tap 4.0 deep (flange 8)	109.484	18.479	11.170
BASE	W wall	M3 tap 5.0 deep (earth point)	0.000	38.549	2.744
LID	through	d2.9 clearance	106.900	2.500	17.220
LID	through	d2.9 clearance	101.200	2.500	17.220
LID	through	d2.9 clearance	90.400	2.500	17.220
LID	through	d2.9 clearance	84.700	2.500	17.220
LID	through	d2.9 clearance	74.200	2.500	17.220
LID	through	d2.9 clearance	62.200	2.500	17.220
LID	through	d2.9 clearance	50.200	2.500	17.220
LID	through	d2.9 clearance	38.500	2.500	17.220
LID	through	d2.9 clearance	26.500	2.500	17.220
LID	through	d2.9 clearance	14.500	2.500	17.220
LID	through	d2.9 clearance	2.500	2.500	17.220
LID	through	d2.9 clearance	106.984	134.500	17.220
LID	through	d2.9 clearance	106.984	114.300	17.220
LID	through	d2.9 clearance	106.984	83.800	17.220
LID	through	d2.9 clearance	106.984	53.300	17.220
LID	through	d2.9 clearance	106.984	22.800	17.220
LID	through	d2.9 clearance	2.584	134.598	17.220
LID	through	d2.9 clearance	8.284	134.598	17.220
LID	through	d2.9 clearance	19.084	134.598	17.220
LID	through	d2.9 clearance	24.784	134.598	17.220
LID	through	d2.9 clearance	35.284	134.598	17.220
LID	through	d2.9 clearance	47.284	134.598	17.220
LID	through	d2.9 clearance	59.284	134.598	17.220
LID	through	d2.9 clearance	70.984	134.598	17.220
LID	through	d2.9 clearance	82.984	134.598	17.220
LID	through	d2.9 clearance	94.984	134.598	17.220
LID	through	d2.9 clearance	2.500	15.098	17.220
LID	through	d2.9 clearance	2.500	30.398	17.220
LID	through	d2.9 clearance	2.500	45.598	17.220
LID	through	d2.9 clearance	2.500	60.898	17.220
LID	through	d2.9 clearance	2.500	76.198	17.220
LID	through	d2.9 clearance	2.500	86.598	17.220
LID	through	d2.9 clearance	2.500	98.598	17.220
LID	through	d2.9 clearance	2.500	110.598	17.220
LID	through	d2.9 clearance	2.500	122.598	17.220
LID	through	d2 H7 (dowel)	14.500	134.598	17.220
LID	through	d2 H7 x 3 slot, along the dowel line	94.984	2.500	17.220
LID	through	M3 tap (jacking screw)	54.742	134.598	17.220
LID	through	M3 tap (jacking screw)	54.742	2.500	17.220
LID	cage face	d5.2 pocket, 2.25 deep from the cage face	9.102	76.299	6.570
LID	cage face	d5.2 pocket, 2.25 deep from the cage face	9.102	60.899	6.570
LID	cage face	d5.2 pocket, 2.25 deep from the cage face	100.302	115.299	6.570
LID	cage face	d5.2 pocket, 2.25 deep from the cage face	100.302	84.499	6.570
LID	cage face	d5.2 pocket, 2.25 deep from the cage face	100.302	53.699	6.570
LID	cage face	d5.2 pocket, 2.25 deep from the cage face	100.302	22.899	6.570
LID	cage face	d5.2 pocket, 2.25 deep from the cage face	7.902	7.899	6.570
LID	cage face	d5.2 pocket, 2.25 deep from the cage face	7.902	129.099	6.570
LID	cage face	d5.2 pocket, 2.25 deep from the cage face	52.102	108.699	6.570
LID	cage face	d5.2 pocket, 2.25 deep from the cage face	51.902	28.499	6.570
LID	cage face	d5.2 pocket, 2.25 deep from the cage face	61.702	68.499	6.570
LID	cage face	d5.2 pocket, 2.25 deep from the cage face	74.102	7.899	6.570
LID	cage face	d5.2 pocket, 2.25 deep from the cage face	22.502	34.099	6.570
LID	cage face	d5.2 pocket, 2.25 deep from the cage face	22.702	103.099	6.570
LID	cage face	d5.2 pocket, 2.25 deep from the cage face	73.902	129.099	6.570

BEL - CRL

ANTENNA / RFM LAB

X-BAND 8-WAY MSL POWER DIVIDER V3

HOLE TABLE (base + lid)

PART	DWG	REV	SHEET
CRL-XB-PD8-03-01/-02	CRL-XB-PD8-03-DWG	V3	4 / 4
DATE	UNITS	PROJ.	TOL.
05-11-2026	mm	1st angle	ISO 2768-mK